

NewsBot Intelligence System 2.0

Advanced NLP Integration and Analysis Platform

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github.com/Tiskalani/ITAI2373-Portfolio

The problem, and what NewsBot does

The problem

Organizations are buried in news. Communications and analyst teams need to know within minutes what is being said about them, whether the tone is turning, and who is being named.

Reading feeds by hand does not scale. Keyword alerts miss context: they cannot tell a positive story from a negative one, cannot group related coverage, and cannot answer a plain question.

The solution

NewsBot 2.0 turns raw article text into structured intelligence:

- category with a confidence score
- sentiment and dominant emotion
- named people, organizations, and places
- a short summary
- topics across a collection, and search by meaning
- answers to plain-language questions

All of it reachable from a web dashboard.

From midterm to final

Midterm

One long Jupyter notebook.

- preprocessing to TF-IDF to classification
- sentiment, NER, LDA topics
- 97.5% test accuracy on BBC News
- a Gradio demo

Worked well, but hard to reuse or extend.

Final

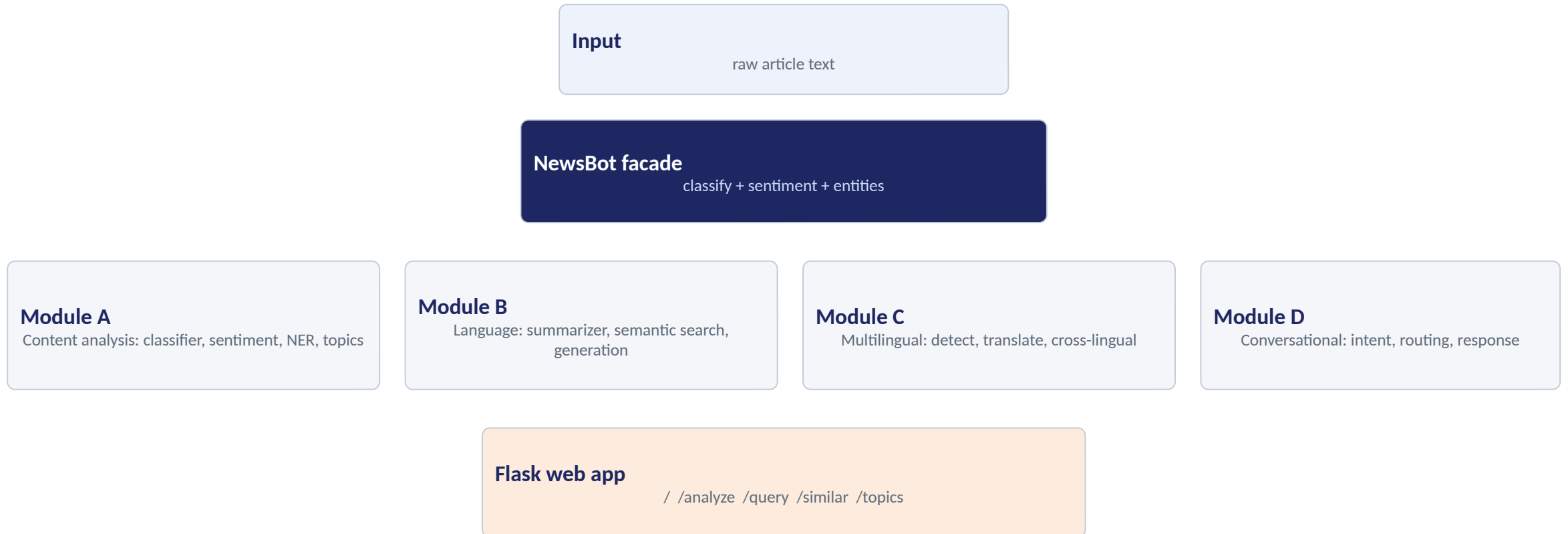
A production-style Python package.

- every capability a small class under `src/`
- four new modules: content analysis, language, multilingual, conversational
- a Flask web app (bonus)
- an automated pytest suite

The refactor is what made the new modules possible.

System architecture

Parse and vectorize once, then reuse



The classifier owns the TF-IDF vectorizer; search and topic modeling keep their own, tuned for recall.

Module A — Advanced content analysis

Classification with an uncertainty guard

Category, confidence, runner-up, and how many words were recognized. Returns 'uncertain' on empty or out-of-scope text instead of guessing. Fixes the midterm bug where short input was labeled sport.

Sentiment and emotion

VADER compound plus TextBlob polarity and subjectivity, and a transparent 8-emotion lexicon (joy, anger, fear, sadness, and more).

Named entity recognition

spaCy entities with an optional domain rule layer that corrects names like Nvidia, OpenAI, and Manchester United.

Topic modeling: LDA and NMF

Unsupervised topics over the corpus, with per-topic top words and a visualization. The topics rediscover the five categories with no labels.

Module B — Language understanding and generation

Summarization

Extractive by default: sentences ranked by TF-IDF salience, top ones returned in order. An abstractive DistilBART path lazy-loads only on request.

Semantic search

Ranks the corpus by cosine similarity to a query, so you search by meaning, not keywords. Optional dense SBERT backend.

Content enhancement

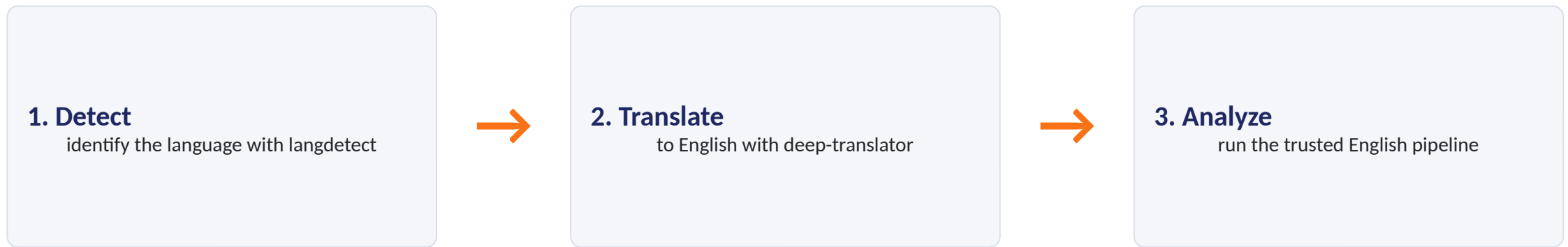
Turns an analysis result into a readable narrative brief for a non-technical reader.

Query expansion

Adds WordNet synonyms to widen a search and catch related coverage.

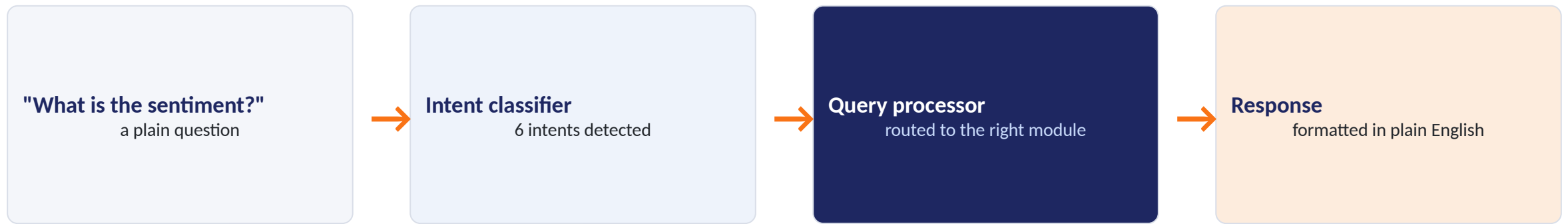
Theme: a light default that runs anywhere, with a heavier transformer upgrade one flag away.

Module C — Multilingual intelligence



A Spanish or French article gets the same classification, sentiment, and entities as an English one. Translating into English is the honest choice with these tools, rather than pretending English-trained models read every language.

Module D — Conversational interface



Intents: classify, sentiment, entities, summarize, search, topics. This only works because each capability is a clean class the router can call. It is the piece that ties the whole system together.

Web application (bonus frontend)

Analyze an article

Paste text, get category and confidence, sentiment and emotion, entities, key terms, and a summary.

Ask NewsBot

Type a question about the article in plain English and get a routed answer.

Find similar articles

Enter a phrase and get the closest coverage by meaning, ranked by similarity.

Same module code, exposed in a browser so a non-technical user can use the system. Flask + HTML/CSS/JS.

Live demo — four fresh articles

Article	Predicted	Confidence	Sentiment / emotion
AI system launch	tech	0.91	positive / neutral
Nasdaq and Apple selloff	business	0.85	positive / neutral
World Cup and Stanley Cup	sport	0.96	positive / joy
UK prime minister resigns	politics	0.97	positive / neutral

All four correct with high confidence. Short off-topic input returns 'uncertain' rather than a forced label.

Discovered topics (LDA, no labels)

0

Topic 0 — Sport game, player, england, time, first, year

1

Topic 1 — Entertainment film, best, award, show, star, year

2

Topic 2 — Tech / mobile mobile, technology, phone, service, people

3

Topic 3 — Business / markets company, market, sale, firm, tax, year

4

Topic 4 — Politics government, labour, party, election, people

Classification results

Model	CV accuracy	Test accuracy	Macro-F1
Linear SVM	0.972	0.975	0.975
Logistic Regression	0.968	0.970	0.969
Random Forest	0.951	0.948	0.947
Naive Bayes	0.945	0.943	0.942

97.5% test accuracy, macro-F1 0.975. The only real confusions are politics and business articles that share vocabulary such as tax and economy.

Business value and use cases

Corporate communications

Monitor brand mentions and alert when sentiment turns negative, the moment that matters most.

Investor relations and finance

Track how a company or sector is covered and surface the entities driving a story.

Media and research teams

Cluster a day's coverage into topics and pull the most relevant articles on demand.

Multilingual monitoring

Detect and translate non-English coverage into the same analysis pipeline.

Honest limitations



Trained on 2004-2005 BBC news, so it is weaker on messy present-day text.



Lexicon sentiment misreads financial framing and can score a profit warning as positive.



The small spaCy model makes occasional NER errors on modern names.



Translation quality depends on a free backend and a network connection.

Stating limits plainly is the point. Trustworthy tooling depends on knowing where it is wrong.

Future work

- Replace lexicon sentiment with a finance-tuned model such as FinBERT
- Fine-tune the entity recognizer on recent news
- Add topic modeling over time to track themes
- Deploy the web app to a public URL, add batch upload and caching
- Explore bias detection and simple fact-checking

Thank you.